Supplementary Appendix: Methodological, Statistical, and Empirical Foundations

Accompanying: Beyond Random Splits: Leakage, Imbalance Handling, Calibration, and Temporal Generalization in BRFSS-Based Heart Disease Classification

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APPENDIX A: COMPLETE FEATURE CROSSWALK AND HARMONIZATION SPECIFICATION

The Behavioral Risk Factor Surveillance System (BRFSS) is administered annually across all 50 U.S. states, the District of Columbia, and three U.S. territories by state health departments in collaboration with the Centers for Disease Control and Prevention (CDC).

The table below provides the full cross-wave mapping for all 17 candidate predictor variables and the primary outcome across the 2020, 2021, and 2022 survey cycles.

Variable Name	CDC 2020 Name	CDC 2021 Name	CDC 2022 Name	CDC Core Questionnaire Text	Response Code Mapping	Preprocessing & Harmonization
_MICHD (Target)	`_MICHD` (`HeartDisease`)	`_MICHD`	`_MICHD`	Calculated: Ever told had coronary heart disease (angina) or myocardial infarction	`1` = Yes (Positive); `2` = No (Negative)	Target binary status indicator. Removed missing / refused.
BMI	`_BMI5`	`_BMI5`	`_BMI5`	Body Mass Index (calculated from reported height and weight)	Continuous integer (implied 2 decimal places: e.g., 2750 = 27.50)	Divided by 100.0. Scaled via training-fit `StandardScaler`.
Smoking	`SMOKE100`	`SMOKE100`	`SMOKE100`	"Have you smoked at least 100 cigarettes in your entire life?"	`1` = Yes; `2` = No	Binary indicator: `Yes` / `No`. One-hot encoded.
AlcoholDrinking	`_RFDRHV7`	`_RFDRHV7`	`_RFDRHV8`	Heavy drinkers (adult men >14 drinks/wk, women >7 drinks/wk)	`1` = No; `2` = Yes	Binary indicator: `No` / `Yes`. (2022 renamed to `_RFDRHV8` due to ALCDAY4 update; logic identical).
Stroke	`CVDSTRK3`	`CVDSTRK3`	`CVDSTRK3`	"(Ever told) you had a stroke?"	`1` = Yes; `2` = No	Binary indicator: `No` / `Yes`. One-hot encoded.
PhysicalHealth	`PHYSHLTH`	`PHYSHLTH`	`PHYSHLTH`	"Now thinking about your physical health, for how many days during the past 30 days was your physical health not good?"	`1-30` = Number of days; `88` = None (0 days)	Mapped `88` *to 0.0*. Continuous float *[0, 30]*. Scaled via `StandardScaler`.
MentalHealth	`MENTHLTH`	`MENTHLTH`	`MENTHLTH`	"Now thinking about your mental health, for how many days during the past 30 days was your mental health not good?"	`1-30` = Number of days; `88` = None (0 days)	Mapped `88` *to 0.0*. Continuous float *[0, 30]*. Scaled via `StandardScaler`.
DiffWalking	`DIFFWALK`	`DIFFWALK`	`DIFFWALK`	"Do you have serious difficulty walking or climbing stairs?"	`1` = Yes; `2` = No	Binary indicator: `No` / `Yes`. One-hot encoded.
Sex	`_SEX` / `SEXVAR`	`_SEX`	`_SEX`	Biological sex recorded from respondent interview	`1` = Male; `2` = Female	Binary category: `Male` / `Female`. One-hot encoded.
AgeCategory	`_AGEG5YR`	`_AGEG5YR`	`_AGEG5YR`	Fourteen-level calculated five-year age categories	`1` = 18-24 ... `13` = 80+, `14` = DK/Refused	Categorical strings ('18-24', ..., '80 or older'). One-hot encoded.
Race	`_IMPRACE`	`_IMPRACE`	`_IMPRACE`	Six-level CDC imputed race/ethnicity variable	`1` = White, `2` = Black, `3` = Asian, `4` = AI/AN, `5` = Hispanic, `6` = Other	Nominal categorical (6 levels). One-hot encoded.
Diabetic	`DIABETE4`	`DIABETE4`	`DIABETE4`	"(Ever told) you have diabetes?"	`1` = Yes; `2` = Yes (pregnancy); `3` = No; `4` = Borderline	Categorical (4 levels: `Yes`, `No`, `No borderline`, `Yes (pregnancy)`). One-hot encoded.
PhysicalActivity	`_TOTINDA`	`_TOTINDA`	`_TOTINDA`	Adults who reported doing physical activity or exercises other than job	`1` = Yes; `2` = No	Binary indicator: `Yes` / `No`. One-hot encoded.
GenHealth	`GENHLTH`	`GENHLTH`	`GENHLTH`	"Would you say that in general your health is: Excellent, Very good, Good, Fair, Poor"	`1` = Excellent, `2` = Very good, `3` = Good, `4` = Fair, `5` = Poor	Ordinal categorical string levels. One-hot encoded.
SleepTime	`SLEPTIM1`	*Rotated Off*	`SLEPTIM1`	"On average, how many hours of sleep do you get in a 24-hour period?"	`1-24` = Hours of sleep	Continuous integer *[1, 24]*. In 2021: median-imputed to *7.0* (sensitivity: NaN routing). In 2022: empirical. Scaled via `StandardScaler`.
Asthma	`ASTHMA3`	`ASTHMA3`	`ASTHMA3`	"(Ever told) you had asthma?"	`1` = Yes; `2` = No	Binary indicator: `No` / `Yes`. One-hot encoded.
KidneyDisease	`CHCKDNY2`	`CHCKDNY2`	`CHCKDNY2`	"(Ever told) you have kidney disease (excluding kidney stones or bladder infection)?"	`1` = Yes; `2` = No	Binary indicator: `No` / `Yes`. One-hot encoded.

Variable Name	CDC 2020 Name	CDC 2021 Name	CDC 2022 Name	CDC Core Questionnaire Text	Response Code Mapping	Preprocessing & Harmonization
SkinCancer	`CHCSCNCR`	`CHCSCNCR`	`CHCSCNC1`	"(Ever told) you had skin cancer?"	`1` = Yes; `2` = No	Binary indicator: `No` / `Yes`. (2022 renamed `CHCSCNC1` due to section order; identical wording).

APPENDIX B: MATHEMATICAL FORMULATION OF COMPLEX SURVEY WEIGHTING

The BRFSS employs a multistage, stratified probability sampling design. Simple random sampling (SRS) variance estimation formulas underestimate true sampling variance due to intra-cluster correlation and unequal probability of selection.

1. Complex Survey Design Parameters

For respondent i belonging to stratum h ($h = 1, \dots, H$) and primary sampling unit (PSU / cluster) j ($j = 1, \dots, n_h$):

- $w_{hij} = \text{_LLCPWT}$: final post-stratification sampling weight produced via iterative proportional fitting (raking) to match state-level Census population margins (age, sex, race, education, marital status, home ownership, telephone ownership).
- Stratum identifier: `_STSTR`.
- Cluster / Primary Sampling Unit identifier: `_PSU`.

2. Taylor Series Linearization for Ratio Estimators

Population-representative performance metrics (such as disease prevalence, sensitivity, and specificity) are nonlinear ratio estimators of the form:

$$\hat{R} = \frac{\hat{Y}}{\hat{X}} = \frac{\sum_{h=1}^H \sum_{j=1}^{n_h} \sum_{i=1}^{m_{hj}} w_{hij} y_{hij}}{\sum_{h=1}^H \sum_{j=1}^{n_h} \sum_{i=1}^{m_{hj}} w_{hij} x_{hij}}$$

For population sensitivity at a fixed threshold τ^*:

- $y_{hij} = I(y_{hij} = 1 \text{ and } \hat{p}_{hij} \ge \tau^*)$
- $x_{hij} = I(y_{hij} = 1)$

For population specificity:

- $y_{hij} = I(y_{hij} = 0 \text{ and } \hat{p}_{hij} < \tau^*)$
- $x_{hij} = I(y_{hij} = 0)$

The first-order Taylor series linearization linearizes the ratio $\hat{R}$ around its expected value:

$$z_{hij} = \frac{y_{hij} - \hat{R} x_{hij}}{\hat{X}}$$

The survey-design variance estimator is then given by:

$$\widehat{\text{Var}}(\hat{R}) = \sum_{h=1}^H \frac{n_h}{n_h - 1} \sum_{j=1}^{n_h} \left(z_{hj\cdot} - \bar{z}_{h\cdot\cdot} \right)^2$$

where:

$$z_{hj\cdot} = \sum_{i=1}^{m_{hj}} w_{hij} z_{hij}, \quad \bar{z}_{h\cdot\cdot} = \frac{1}{n_h} \sum_{j=1}^{n_h} z_{hj\cdot}$$

The design-based standard error is $\text{SE}(\hat{R}) = \sqrt{\widehat{\text{Var}}(\hat{R})}$, and the design effect (DEFF) is defined as:

$$\text{DEFF} = \frac{\widehat{\text{Var}}_{\text{complex}}(\hat{R})}{\widehat{\text{Var}}_{\text{SRS}}(\hat{R})}$$

Empirically, for BRFSS 2021 and 2022, DEFF values range from 1.76 to 4.38, demonstrating that standard unweighted or unstratified bootstrap resampling substantially underestimates true sampling variance.

APPENDIX C: COMPLETE MISSINGNESS SELECTION AUDIT

To evaluate whether complete-case evaluation introduced substantial target selection bias, we forensically audited all excluded respondents in 2021 and 2022.

1. Retention and Attrition Breakdown

Metric / Cohort Property	BRFSS 2021 Wave	BRFSS 2022 Wave
Total Ingested Records	*438,693*	*445,132*
Complete-Case Records Retained	*348,568* (*79.46\%*)	*339,264* (*76.22\%*)
Excluded Records	*90,125* (*20.54\%*)	*105,868* (*23.78\%*)
Missing Target (`_MICHD`)	*4,635* (*1.06\%*)	*5,021* (*1.13\%*)
Missing Continuous/Numeric Features	*59,421* (*13.55\%*)	*66,154* (*14.86\%*)
- *Missing BMI (`_BMI5`)*	*46,852* (*10.68\%*)	*48,806* (*10.96\%*)
- *Missing Physical Health (`PHYSHLTH`)*	*10,488* (*2.39\%*)	*11,947* (*2.68\%*)
- *Missing Mental Health (`MENTHLTH`)*	*8,521* (*1.94\%*)	*9,812* (*2.20\%*)
Missing Categorical Predictors	*51,919* (*11.83\%*)	*67,330* (*15.13\%*)
- *Missing Heavy Drinking (`_RFDRHV`)*	*34,992* (*7.98\%*)	*49,705* (*11.17\%*)
- *Missing Smoking Status (`SMOKE100`)*	*24,461* (*5.58\%*)	*34,959* (*7.85\%*)
- *Missing General Health (`GENHLTH`)*	*1,842* (*0.42\%*)	*2,105* (*0.47\%*)

2. Demographic and Target Distribution: Complete Cases vs. Excluded Respondents

Cohort Characteristic	2021 Included Complete Cases	2021 Excluded Records	2022 Included Complete Cases	2022 Excluded Records
Target Disease Prevalence	*8.16\%*	*8.05\%* (*\Delta = +0.11\%*)	*8.92\%*	*9.42\%* (*\Delta = -0.50\%*)
Elderly (Ages 70+)	*24.53\%*	*24.03\%* (*\Delta = +0.50\%*)	*25.24\%*	*25.14\%* (*\Delta = +0.10\%*)
Female Biological Sex	*51.80\%*	*60.32\%* (*\Delta = -8.52\%*)	*51.24\%*	*58.63\%* (*\Delta = -7.39\%*)
White Race/Ethnicity	*76.74\%*	*71.81\%* (*\Delta = +4.93\%*)	*76.23\%*	*70.74\%* (*\Delta = +5.49\%*)
Black Race/Ethnicity	*7.15\%*	*9.82\%* (*\Delta = -2.67\%*)	*7.34\%*	*9.91\%* (*\Delta = -2.57\%*)
Hispanic Race/Ethnicity	*8.43\%*	*10.84\%* (*\Delta = -2.41\%*)	*8.66\%*	*11.21\%* (*\Delta = -2.55\%*)

Assessment of Bias:

1. *Target Selection Bias:* The difference in self-reported heart disease prevalence between included complete cases and excluded respondents is negligible (*0.11\%* in 2021 and *0.50\%* in 2022), and elderly representation is virtually identical (*\Delta \le 0.50\%*).

2. *Demographic Selection Bias:* Refusal to report body weight and height (driving BMI missingness) and alcohol consumption frequency was disproportionately higher among female respondents (*+7.4\%* to *+8.5\%*) and racial/ethnic minority respondents (*+5.0\%* to *+5.5\%*). Renormalizing post-stratification sampling weights within the complete-case subcohort partially rebalances sample margins against U.S. Census totals, but does not completely cure potential non-ignorable item non-response.

APPENDIX D: 2021 SLEEP DURATION ROTATION AND SENSITIVITY BENCHMARKING

In the BRFSS surveillance system, certain questionnaire modules undergo biennial rotation between odd and even survey years. The sleep duration item (`SLEPTIM1`) was included in the 2020 core survey, rotated off the nationwide core survey in 2021 (appearing only in optional state-added modules), and restored to the nationwide core survey in 2022.

In the 2020 training dataset (*N = 255,836*), `SleepTime` had zero missing values after pre-filtering. Consequently, XGBoost's default tree routing rules for missing values (`np.nan`) were never activated during training and defaulted to split paths associated with higher risk.

We benchmarked two explicit operational strategies on the *N = 348,568* complete-case 2021 cohort:

1. **Primary Predefined Harmonization Rule (Median Imputation):** Sleep duration was imputed with a constant value of *7.0* hours, representing the exact median and modal sleep duration in the 2020 training set.

2. **Secondary Sensitivity Benchmark (Native NaN Tree Routing):** Missing sleep values were passed directly as `np.nan` through the preprocessor pipeline and routed via XGBoost's default branch rules.

Evaluation Metric	Primary: Median Imputation (*7.0* hrs)	Sensitivity: Native Missing Routing (`NaN`)	Absolute Shift (*Delta*)	2022 Empirical Baseline (*100*\% Measured)
ROC-AUC	**0.8407**	**0.8344**	*-0.0063*	**0.8397**
PR-AUC	**0.3304**	**0.3077**	*-0.0227*	**0.3501**
Sensitivity at *tau = 0.0816*	**80.68%** (*22,948* TP)	**90.18%** (*25,651* TP)	*+9.50*\%	**81.96%** (*24,798* TP)
Specificity at *tau = 0.0816*	**72.24%** (*231,245* TN)	**58.74%** (*188,035* TN)	*-13.50*\%	**70.87%** (*218,995* TN)
Positive Predictive Value (PPV)	**20.52%**	**16.26%**	*-4.26*\%	**21.60%**
Brier Score	**0.0634**	**0.0698**	*+0.0064*	**0.0680**
Expected Calibration Error (ECE)	**0.0055**	**0.0514**	*+0.0459* (*9.3**times* error)	**0.0044**
Mean Predicted Probability	**0.0871**	**0.1330**	*+0.0459* (*+52.7*\% drift)	**0.0931**

Conclusion:

Uncalibrated tree routing on unobserved missing features caused an upward probability drift of *+4.6*\% points, artificially inflating recall to *90.18*\% at the expense of a severe collapse in specificity (*58.74*\%) and a *9.3*-fold explosion in calibration error (*\text{ECE} = 0.0514*). In contrast, imputing the 2020 training median (*7.0* hours) maintained pristine probability calibration (*\text{ECE} = 0.0055*) and closely mirrored the 2022 empirical wave results (*81.96*\% recall, *70.87*\% specificity, *\text{ECE} = 0.0044*).